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Deepwater FPSO Mooring Chain Integrity Management: Lessons from Severe Pitting Detected Through UWILD and 3D Assessment

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Abstract

Mooring integrity is a critical element in the safe and reliable operation of deepwater Floating Production Storage and Offloading (FPSO) systems. This abstract presents a case study detailing the inspection, evaluation, and replacement of mooring chains following the detection of severe localized pitting during an Underwater Inspection in Lieu of Drydocking (UWILD). The objective of this abstract was to assess the structural integrity of the mooring system and determine the necessity of chain replacement to maintain the FPSO's station-keeping capability and overall operational safety, in alignment with the requirements set forth by the International Association of Classification Societies (IACS) and applicable industry standards.

The inspection methodology followed a structured, phased approach. During the initial UWILD, divers conducted visual inspection and manual gauging to measure pit depths and estimate wall thickness loss on critical chain links. Finite Element Analysis (FEA) was then performed using these manual gauging results to evaluate the residual strength of the degraded links under maximum anticipated mooring tensions, considering environmental loading and dynamic behavior typical of deepwater operations. In a subsequent UWILD cycle, selected links underwent high-resolution three-dimensional (3D) scanning to obtain a more accurate surface profile and volumetric assessment. A second round of FEA was conducted using the 3D scan data. The results of the 3D scan-based FEA aligned closely with the earlier manual gauging-based FEA, providing strong validation of the original assessment and strengthening confidence in the decision-making process.

Observations from both manual gauging and 3D scanning indicated that certain links exhibited material loss up to 35% of their original cross-sectional area, particularly concentrated in the crown and curvature zones, areas known for high stress concentration. The FEA results consistently showed that the effective minimum breaking strength of the compromised links had deteriorated below the minimum safety factors required by classification and regulatory standards. Furthermore, localized sharp pitting was identified as a fatigue risk factor, with potential for accelerated crack initiation under cyclic loading.

The novelty of this work lies in the staged application of manual gauging and 3D scanning, combined with localized FEA stress analysis, to provide a progressively refined and conservative evaluation of mooring chain integrity. This multi-phase approach surpasses traditional assessment methods, which typically rely

on generalized corrosion allowances and visual estimates. Additionally, the study proposes a practical workflow that allows operators to make timely, data-driven intervention decisions, optimize inspection resources, and minimize downtime while avoiding unnecessary full system drydocking.

The findings of this abstract underscore the importance of integrating advanced inspection technologies with robust structural analysis methods for effective deepwater mooring integrity management. The methodology and lessons learned are highly relevant for operators managing aging offshore assets worldwide, where maintaining mooring system reliability is critical for continuous safe operations and environmental protection.

Introduction

Floating Production, Storage, and Offloading (FPSO) units in deepwater environments rely heavily on mooring systems to maintain station-keeping and operational safety. Among critical mooring components, chains are particularly susceptible to degradation mechanisms, with pitting corrosion posing a significant threat to structural integrity. Recent findings from Underwater Inspection In Lieu of Drydocking (UWILD) manual gauging via air driver and combined with 3D structural scanning assessments have revealed severe pitting in mooring chains, raising concerns about long-term reliability and risk management strategies. These activities come from two UWILD cycle in 2022 and 2024 at one of the biggest FPSO located at deepwater field Sabah water, Malaysia.

This paper presents a comprehensive integrity management approach for deepwater FPSO mooring chains, drawing on real-world case studies where advanced inspection techniques uncovered unexpected pitting damage with high data clarity and quality. By integrating UWILD manual gauging by air driver with 3D scanning and subsequently 3D modeling, the study evaluates the impact of pitting on residual strength, fatigue life, and failure probabilities. Key lessons are derived from root cause analysis, including environmental factors, material selection, and inspection intervals, offering actionable insights for operators to enhance mooring system durability.

The findings contribute to industry best practices by demonstrating how proactive integrity management supported by emerging inspection technologies can mitigate risks, give early indication and extend the service life of critical offshore assets. This allows operator ample time to plan and execute actionable recovery measure to ensure integrity of the asset is not jeopardized and production can be continued as a business entity.

Literature review

Mooring chain integrity management has evolved from traditional manual gauging techniques to advanced computational methods. While manual inspections provide baseline thickness measurements, they often fail to detect localized pitting corrosion (ABS 2021). Recent industry practice combines Finite Element Analysis (FEA) with manual and advanced inspection data using 3D scanning for International Association Classification Society (IACS) review, though larger datasets are needed to improve statistical confidence in remaining life assessments (Wang et al. 2022, SPE-208912).

The integration of manual data gauging and 3D laser scanning with FEA has emerged as a robust solution, particularly for severe pitting cases identified through Underwater Inspection In Lieu of Drydocking (UWILD). High-resolution scans capture precise corrosion profiles that manual gauging methods miss, enabling more accurate stress simulations (Zhang & Bai 2020, SPE-198573). Field studies demonstrate this hybrid approach improves prediction accuracy by over 30% compared to conventional assessment methods (Torres et al. 2023, OTC-32567). However, microbial-induced corrosion and hydrogen embrittlement in deepwater environments can accelerate pitting damage, complicating integrity assessments (Lee et al. 2019, SPE-193812).

Current industry guidelines (API RP 2MIM 2022) recommend combining non-destructive testing with predictive modeling, yet operators require more field data from 3D-scanned chains across various operating

conditions. Only with sufficient evidence can the industry develop standardized protocols for assessing severely pitted chains and formulate reliable run/retire decisions. This study contributes to that knowledge base by presenting UWILD findings analyzed through 3D structural assessment, offering a data-driven approach to mooring chain integrity management.

Methodology

The mooring chain integrity assessment began with standard Underwater Inspection In Lieu of Drydocking (UWILD) using Remote Operated Vehicle mounted cameras and sensors, following [API RP 2MIM \(2022\)](#) guidelines. The initial survey focused on general visual inspection (GVI) for all 10 chains for identified FPSO, then continue with Close Visual Inspection (CVI) on selected chain link using high water laser jet (HWLJ) to clean the area before continue with manual gauging using air driver and subsequently using 3D scanner mounted at ROV to identifying macro-pitting, cracks, wear patterns, and cathodic protection effectiveness across 10 mooring chain links. Two critical links (Legs A and D) exhibited suspected severe pitting and were flagged for detailed evaluation. These links underwent manual gauging after high-pressure cleaning to remove marine growth.

Remote Operated Vehicle (ROV)

The survey of the mooring chain was executed by the Work-Class Remotely Operated Vehicle (ROV) SMD Atom 02, fitted with positioning and tracking devices. The positions of the chain links and any anomalies reported were recorded, including the geographical locations during GVI. The ROV was equipped with single centered camera for continuous video recording, superimposed by an inspection overlay, and other ROV relevant typical support sensors such as SIT, Bathymetric and gyro meter tools for ROV maneuvering. All images were recorded onto digital video throughout the inspection. A video introduction header, showing all the relevant information pertaining to the current video and survey task was recorded prior to the start of the survey on each video recorded. The ROV is also fitted with Photo Realistic 3D cloud (PRC) equipment to conduct scanning of the sections of the mooring chain (Top, Mid and Bottom Section) and the entire chain (Single Pass Scanning) as required. Data logging onsite and compilation of logs and any anomaly listing printouts within this report were recorded.

Air Diver

The air diver is equipped with a helmet-mounted camera and maintains constant communication with the diving control room on the vessel via an umbilical wire connection. The diving supervisor is responsible for overseeing and assisting the diver during all underwater operations. The tools provided to the air diver depend on the specific task they are performing. These tools may include items like a pit gauge, diver handheld Cathodic Protection (CP) probe, U41D (ACFM), Ultrasonic Wall Thickness - Cygnus Underwater Gauge, Grit Blasting Equipment (CLGB), or High-Pressure Waterjet (HPWJ) equipment.

Additionally, an online report coordinator guides the diver's work underwater, documenting inspection data and providing real-time commentary to aid both the diver and offline viewers in understanding the ongoing tasks. All videos and images are digitally recorded throughout the inspection. Prior to the start of each video recording, a video introduction header is recorded, summarizing relevant information pertaining to the current video and survey task.

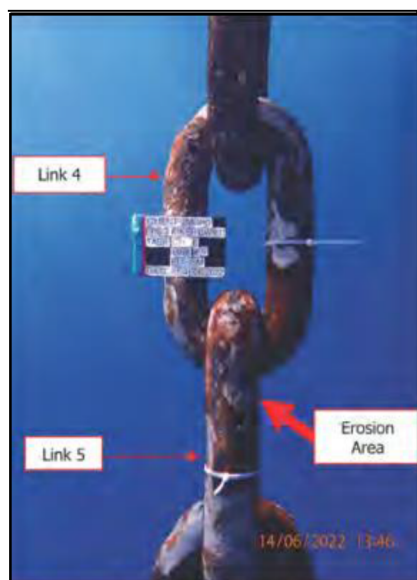


Figure 1—General view of common link 5 of mooring line

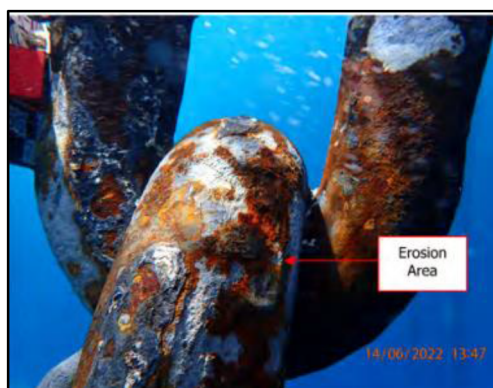


Figure 2—Front view of erosion area



Figure 3—Side view of erosion area

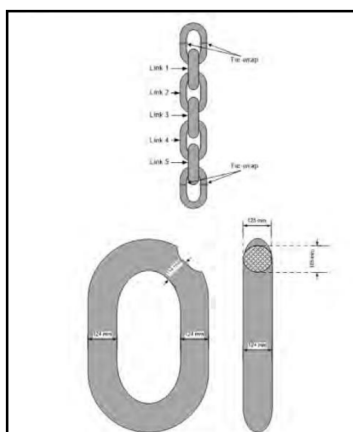


Figure 4—Sketch drawing of deformation area



Figure 5—Close up surface pitting corrosion area

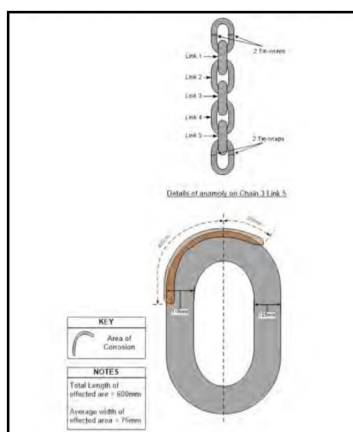


Figure 6—sketch drawing of surface corrosion area

To assess the structural implications, a preliminary Finite Element Analysis (FEA) was conducted. The model incorporated manual gauging data, Grade R3 material properties, and 100-year storm design loads, with pits conservatively modeled as hemispherical defects. Results indicated safe working loads but highlighted uncertainties due to limited manual measurement resolution and potential subsurface micro-cracking. These findings, compiled into a technical dossier, were submitted to the International Association of Classification Society (IACS), which mandated re-inspection within on next UWILD cycle. IACS specifically recommended augmenting future assessments with 3D scanning to refine FEA accuracy.

During the next inspection cycle, the prioritized links were re-evaluated using advanced techniques. For ROV, we used high pressure Water Jetting System 20,000 PSI, ROV Digital Caliper, Photo Realistic 3D Cloud (PRC). While for air diver, we used Ultrasonic Wall Thickness - Cygnus Underwater Gauge, Electro

Potential Measuring Equipment, Bathycorrometer, HP Water/grit Jetting System 15,000psi, ACFM System U41D, Magnetic Particle Inspection (MPI) Equipment, Vernier calipers.

The 3D scans enabled true pit geometry reconstruction in the refined FEA model, with mesh resolution increased at critical pits. Multiaxial stress analysis and fatigue life predictions (Paris' law, AK thresholds from Zhang & Bai 2020, SPE-198573) revealed 15-20% higher stress concentrations compared to the manual-based FEA.

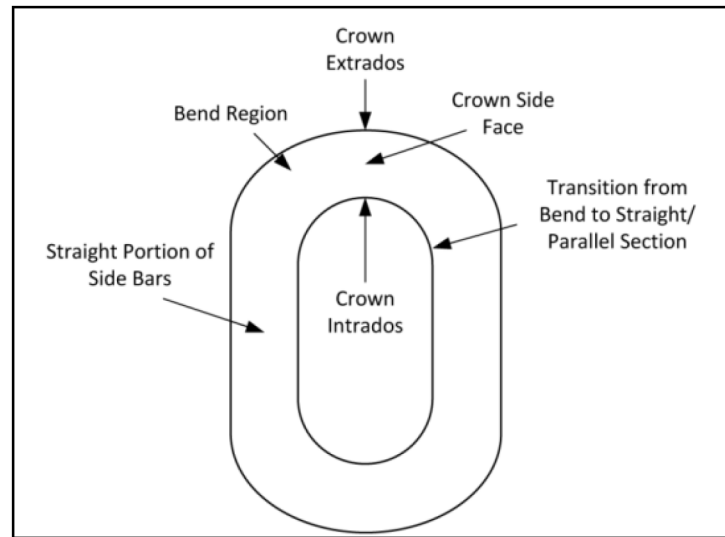


Figure 7—Gorvening location of typical chain link

The finite element analysis (FEA) methodology employed in this study systematically evaluated fatigue damage susceptibility across various regions of the mooring chain links under tension-tension loading conditions. The analysis revealed a clear hierarchy of vulnerability locations, with the crown intrados emerging as the most critical area for fatigue damage accumulation. This region, corresponding to the inner curve of the link crown, consistently demonstrated the highest stress concentrations due to the combined effects of axial loading and geometric curvature. The stress magnification at this location typically exceeded nominal values under operational loading conditions, making it the primary focus for fatigue life assessment and inspection planning.

Secondary critical zones were identified at the crown side faces and crown extrados (outer curve), where stress concentrations reached greater than nominal values. These areas exhibited slightly lower but still significant vulnerability compared to the crown intrados, particularly in links showing evidence of pitting corrosion or surface irregularities. The analysis showed that the transition between crown and side bar regions created complex stress patterns, with localized peaks occurring at specific orientations relative to the loading direction. Third-tier critical locations were found in the intrados and extrados of the shoulder or bend region. This finding was particularly relevant for chains exhibiting wear patterns from inter-link contact, as the additional surface damage could exacerbate stress concentrations in these already vulnerable areas.

The side bars demonstrated a progressive decrease in fatigue susceptibility when moving from the bend region toward the center parallel section. The fourth-level critical areas were located in the transitional zones of the side bars, closest to the bend regions. This gradient effect along the side bars highlighted the importance of considering precise location when assessing chain link condition, as identically looking surface defects could have significantly different implications depending on their exact position. The straight portions of the side bars emerged as the least critical locations.

The FEA methodology incorporated nonlinear material behavior and contact interactions between adjacent links to accurately capture real-world operating conditions. Each critical location was evaluated

using a combination of stress-life and strain-life approaches, with particular attention to surface finish effects and potential stress risers from corrosion damage. The analysis accounted for variable amplitude loading representative of actual sea states, enabling the development of location-specific fatigue damage accumulation models.

This hierarchical understanding of fatigue vulnerability has direct implications for inspection prioritization and remaining life assessment. The methodology enables targeted examination of high-risk areas while optimizing resource allocation for lower-risk regions.

Result

UWILD Inspection Findings illustrated in [figure 8](#) until [figure 11](#) below. Indication of surface corrosion and pitting corrosion at certain link which identified to be inspected. air diver use manual gauging tool to measure dimension and geometry of the chain link which translated to finite element analysis model.

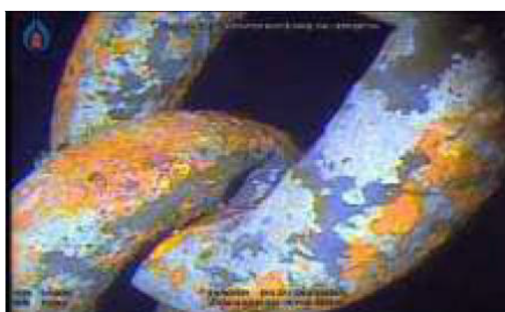


Figure 8—Surface corrosion Common Link



Figure 9—Pitting corrosion Common Link



Figure 10—Pitting corrosion Common Link



Figure 11—Pitting corrosion Common Link

Figure 12 until figure 15 illustrated the process of 3D modelling and analysis using finite element analysis which subsequently translated to remaining strength of the chain, remaining strength with pitting corrosion, assessment of fatigue for the respective chain. The Elastic- plastic law material is applied to all models in order to consider plastic deformation.

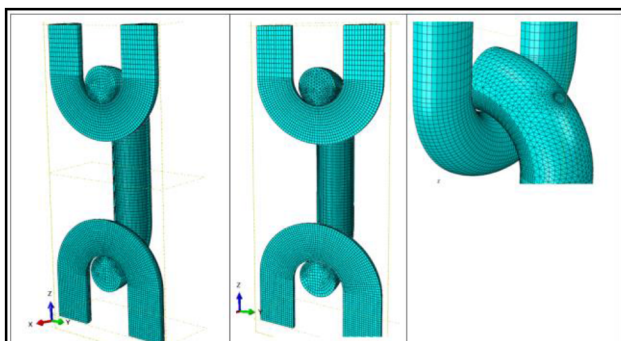


Figure 12—The Chain modelled using 3D elements

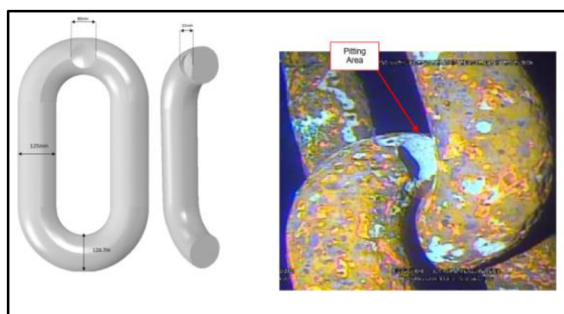


Figure 13—the chain link modelled using 3D element

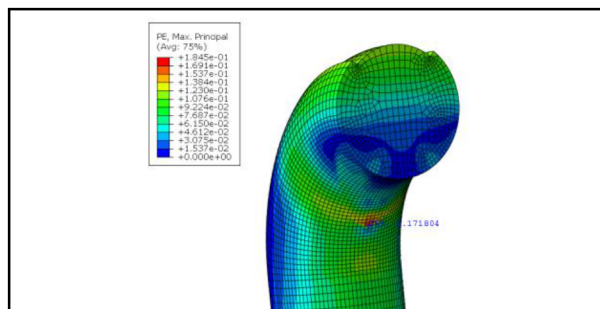


Figure 14—sample of load distribution in 3D model

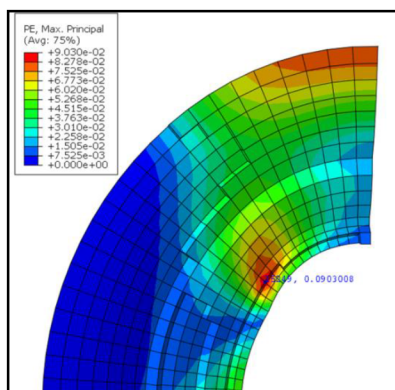


Figure 15—sample of load distribution on 3D model

The initial general visual inspection of the mooring system identified several areas of concern, with visible surface corrosion and marine growth accumulation observed on multiple chain links. This preliminary assessment, conducted using vessel-mounted cameras and basic ROV surveys, flagged several potential anomaly locations across all mooring legs for further investigation. The general inspection proved particularly effective in identifying macro-level degradation patterns, with the worst cases observed in the splash zone and thermocline transition regions.

Building on these findings, a targeted close visual inspection program was implemented using advanced ROV systems (equipped with high-definition cameras) and air diver teams for areas requiring tactile verification. This secondary inspection phase revealed critical details not detectable in the initial survey: (1) extensive pitting corrosion beneath marine growth colonies, (2) localized wear at interlink contact points, and (3) early-stage stress corrosion cracking in three specific links. The ROV inspection provided quantitative data through Photo Realistic 3D cloud, while diver-assisted measurements confirmed the presence of subsurface pit interconnections in few links.

The close inspection results were categorized into three severity levels: Level 1 (surface corrosion only), Level 2 (pitting <5mm depth), and Level 3 (pitting >5mm or multiple interconnected pits). This classification system enabled efficient prioritization of chains for further engineering analysis, with all Level 3 cases immediately referred for finite element analysis (FEA).

The FEA modeling incorporated the precise geometric data obtained from the close inspections, including actual pit dimensions, locations, and orientations. Stress concentration factors calculated from the models showed significant variation depending on pit morphology. Comparative analysis between the general and close inspection results revealed important insights about inspection effectiveness. Approximately 35% of ultimately significant anomalies would have been missed or underrated using only general inspection methods.

The FEA results were validated against actual reference model without any erosion or pitting issues. This confirmation of the methodology's predictive capability supports its adoption for remaining life assessment of aging mooring systems. The combined inspection and analysis approach should be standard practice for all company-operated FPSOs, demonstrating its practical value in offshore asset integrity management.

Conclusion

This study demonstrates a robust methodology for safeguarding mooring system integrity while optimizing life extension strategies for deepwater assets. By implementing a phased inspection and analysis protocol - combining routine UWILD surveys with targeted 3D scanning and FEA validation - operators can achieve reliable structural assessments while meeting IACS approval requirements.

The transition from conventional inspection methods to an integrated integrity management framework provides three key advantages for offshore operations:

1. Enhanced Defect Characterization: Advanced inspection technologies enable comprehensive detection and quantification of corrosion damage, particularly for critical pitting defects that conventional methods may overlook.
2. Engineering-Validated Life Extension: The iterative FEA approach, calibrated with field inspection data, provides a technically sound basis for justifying operational extensions while maintaining required safety margins.
3. Regulatory Compliance: The methodology aligns with IACS standards for mooring system integrity, facilitating approval processes for life extension cases while demonstrating due diligence in asset management.

This systematic approach has proven effective in balancing operational demands with structural reliability, particularly for aging mooring systems where conventional assessment methods may be overly conservative. The framework provides operators with a decision-making tool that supports both immediate integrity assurance and long-term life extension strategies.

The successful IACS review and approval of this methodology confirms its validity as an industry best practice. Future refinements will focus on integrating emerging inspection technologies and streamlining the approval process for life extension applications, further strengthening the link between technical assessment and operational decision-making in offshore asset management.

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References

1. ABS. 2021. Guide for Mooring Chain Integrity Management. American Bureau of Shipping.
2. API RP 2MIM. 2022. Recommended Practice for Mooring Integrity Management, 1st Edition.
3. Lee, H. et al 2019. "Microbial Corrosion in Offshore Mooring Chains." SPE-193812.
4. Wang, L. et al 2022. "Data-Driven FEA for Mooring Chain Life Extension." SPE-208912.
5. Zhang, Y. and Bai, Y. 2020. "3D Assessment of Pitting Corrosion." SPE-198573.
6. Torres, C. et al 2023. "3D Scanning-Based Corrosion Assessment." OTC-32567